Codeforces and Polygon may be unavailable from February 8, 16:00 (UTC) to February 9, 07:00 (UTC) due to technical maintenance. ×

E. Enemy is weak

time limit per test: 5 seconds
memory limit per test: 256 megabytes

The Romans have attacked again. This time they are much more than the Persians but Shapur is ready to defeat them. He says: "A lion is never afraid of a hundred sheep".

Nevertheless Shapur has to find weaknesses in the Roman army to defeat them. So he gives the army a weakness number.

In Shapur's opinion the weakness of an army is equal to the number of triplets i, j, k such that $i < j < k$ and $a_i > a_j > a_k$ where a_x is the power of man standing at position x. The Roman army has one special trait — powers of all the people in it are distinct.

Help Shapur find out how weak the Romans are.

Input

The first line of input contains a single number n ($3 \le n \le 10^6$) — the number of men in Roman army. Next line contains n different positive integers a_i ($1 \le i \le n, 1 \le a_i \le 10^9$) — powers of men in the Roman army.

Output

A single integer number, the weakness of the Roman army.

Please, do not use `%lld` specificator to read or write 64-bit integers in C++. It is preffered to use `cout` (also you may use `%I64d`).

Examples

input	Copy
3 3 2 1	

output	Copy
1	

input	Copy
3 2 3 1	

output	Copy
0	

input	Copy
4 10 8 3 1	

output	Copy
4	

input	Copy
4 1 5 4 3	

output	Copy
1	

Codeforces Beta Round 57 (Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
361211746	Feb/03/2026 03:31	Accepted
361211230	Feb/03/2026 03:17	Time limit exceeded on test 25
361210860	Feb/03/2026 03:05	Memory limit exceeded on test 21
361210836	Feb/03/2026 03:04	Memory limit exceeded on test 21
361210525	Feb/03/2026 02:52	Accepted

→ Problem tags

data structures trees *1900

No tag edit access

→ Contest materials

- Announcement (oth)
- Round 57-E (en)
- Round 57-B (en)
- Round 57-D (en)
- Round 57-C (en)